

PROJECT RESEARCH REPORT

An Intelligent Framework for Prediction and Avoidance of Cyberattack with Digital Evidence Analysis

Project Overview, Literature Review (2020–2026), Research Gap Analysis and Proposed Novelty

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Abstract

Cyberattacks increasingly unfold as multi-stage campaigns in which attackers move from initial access to privilege escalation, lateral movement and finally exfiltration or impact. Conventional digital forensics analyses evidence only after damage has occurred, while intrusion detection systems raise alerts without anticipating the attacker’s next move. This project proposes an intelligent framework that uses digital evidence — system logs, provenance graphs, network flows and memory artefacts — to predict the next stage of an ongoing attack and to trigger avoidance actions before that stage is reached. A review of literature from 2020 to 2026 across IEEE, Springer, Elsevier, arXiv and Google Scholar shows that forensic investigation, attack-stage prediction and automated response have matured as separate research streams, but no framework unifies them while accounting for the forensic consequences of defensive actions. The proposed novelty, Evidence-Preserving Pre-emptive Containment (EPPC), introduces an Evidence Loss Cost model, dual-objective action selection, evidence-grounded prediction and an evidence-degradation robustness protocol, together with a new Evidence Retention Score metric.

Keywords: cyberattack prediction, digital forensics, provenance graph, MITRE ATT&CK, graph neural networks, automated incident response, forensic readiness.

1. Introduction

1.1 Background

Modern organisations face persistent and targeted threats, often called Advanced Persistent Threats (APTs), that remain inside networks for extended periods and progress through several stages. Every action an attacker performs leaves traces: process creations, file modifications, registry changes, authentication records and network connections. These traces constitute digital evidence. Traditionally, such evidence is examined by forensic investigators after an incident to reconstruct what happened. This project reverses that direction by using evidence while the attack is still in progress to anticipate and prevent its later stages.

1.2 Problem Statement

Existing security systems suffer from three disconnected limitations. Forensic tools are retrospective and do not prevent damage. Detection systems identify malicious activity but do not forecast the attacker’s subsequent behaviour. Automated response systems block threats but frequently destroy volatile evidence, such as process memory and live network state, which investigators later require. There is therefore a need for a single intelligent framework that analyses digital evidence, predicts upcoming attack stages and selects avoidance actions that stop the attack without eliminating the evidence that supports investigation and attribution.

1.3 Objectives

- To collect and model heterogeneous digital evidence as a temporal provenance graph.
- To predict the next attack stage and technique aligned with the MITRE ATT&CK framework.

- To select avoidance actions that jointly minimise attack progression and forensic evidence loss.
- To ensure every prediction is explainable through the specific evidence supporting it.
- To evaluate robustness when parts of the evidence are missing or deliberately erased.

2. Fundamental Concepts

Concept	Description	Role in This Project
Digital Evidence	Logs, audit records, network flows, memory images and file-system artefacts produced by system activity	Primary input to the framework
Cyber Kill Chain	Lockheed Martin model describing sequential attack phases from reconnaissance to actions on objectives	Defines the progression to be predicted
MITRE ATT&CK	Knowledge base of adversary tactics and techniques, each with a unique identifier	Label space for stage and technique prediction
Provenance Graph	Graph in which nodes are processes, files and sockets and edges are system actions such as read, write, fork and connect	Structured representation of evidence
APT	Stealthy, long-duration, multi-stage attack	Realistic scenario where prediction offers lead time
Forensic Soundness	Preserving evidence integrity and chain of custody	Constraint on avoidance actions
SOAR	Security Orchestration, Automation and Response platforms	Execution layer for avoidance actions

3. System Overview

The framework operates as a closed loop consisting of six layers.

Layer	Function	Output
1. Evidence Acquisition	Collects host audit logs, Sysmon events, network flows and memory snapshots	Raw evidence stream
2. Evidence Modelling	Parses events and builds a time-stamped provenance graph	Temporal evidence graph
3. Detection and Mapping	Identifies suspicious subgraphs and maps them to ATT&CK techniques	Observed attack stages
4. Stage Prediction	Temporal GNN and Transformer forecast the next tactic and technique	Ranked next-stage probabilities
5. Avoidance Decision	Selects the containment action balancing risk and evidence loss	Chosen response action
6. Forensic Reporting	Preserves evidence and generates an explainable incident report	Report and evidence package

4. Literature Review (2020–2026)

The literature was examined across four research streams relevant to the project. The review draws on IEEE, Springer, Elsevier, USENIX, NDSS, PLOS One and arXiv publications.

4.1 Forensic Analysis and Attack Investigation

Work	Venue / Year	Contribution	Limitation
Qadir & Varol – Role of ML in Digital Forensics	IEEE ISDFS 2020	Reviews machine learning applications in forensic analysis	Descriptive; no framework
UNICORN	NDSS 2020	Graph-sketch based anomaly detection on provenance data	Detection only; no prediction
ATLAS	USENIX Security 2021	Sequence-based learning to reconstruct attack stories from audit logs	Retrospective investigation
ThreaTrace	IEEE TIFS 2022	Node-level GNN detection on provenance graphs	No stage forecasting
KAIROS	IEEE S&P 2024	Temporal-graph intrusion detection and investigation	No response component
ML-based Criminal Behaviour Analysis	PLOS One 2025	Links behavioural analysis with digital evidence for attribution and prediction	Limited to browser artefacts
Admissibility Framework for Open-Source Tools	PLOS One 2025	Three-phase framework combining forensic processes, validation and readiness under the Daubert Standard	Legal focus; no AI prediction

4.2 Cyberattack and Attack-Stage Prediction

Work	Venue / Year	Contribution	Limitation
Chayal & Patel – Review of ML for Cyberattack Prediction	Springer 2021	Surveys data-mining and ML prediction methods	Dataset-level, not evidence-level
Recent Advancements in ML for Cybercrime Prediction	arXiv 2023	Comprehensive survey of prediction techniques	Survey only
KillChainGraph	arXiv 2025	ATTACK-BERT phase mapping with LightGBM, Transformer, BERT and GNN soft-voting ensemble	Built on ATT&CK knowledge base, not live host evidence; no avoidance
StageFinder	arXiv 2026	GNN encoding of provenance with LSTM temporal modelling; macro F1 of 0.96 on DARPA OpTC and Transparent Computing	Stage estimation only; no action

4.3 LLM and ATT&CK-Driven Investigation

Work	Venue / Year	Contribution	Limitation
OMNISEC	arXiv 2025	LLM with retrieval-augmented behaviour prompting for provenance-based intrusion detection	Detection only
CTINexus	2025	In-context learning to extract entity–relation triplets from threat intelligence reports	Intelligence extraction, not prevention

Work	Venue / Year	Contribution	Limitation
PROVCON	2025	Reproduces campaigns in a cyber range to derive provenance graphs from CTI	Investigation focus
SIEM ATT&CK Mapping (Splunk, Elastic, DeTT&CT)	Industry	Rule-based correlation of events to techniques	Cannot identify unmodelled or emerging techniques

4.4 Automated Response and Autonomous Defence

Work	Venue / Year	Contribution	Limitation
CybORG / CAGE Challenges	2021–2024	Reinforcement learning agents for autonomous network defence	Ignores forensic evidence preservation
AI-Enabled Cloud Incident Detection and Response	arXiv 2024	AI system for detection and response in cloud environments	Response actions not evaluated for evidence impact
SOAR Playbook Automation	Industry	Rule-driven automated containment	Aggressive actions such as re-imaging destroy volatile evidence

5. Research Gap Analysis

ID	Research Gap	Supporting Observation
G1	Forensics is retrospective and prevention is prospective; the two are rarely unified in one framework	Investigation and response literature form separate communities
G2	Prediction studies stop at stage estimation and do not select or execute avoidance actions	StageFinder and KillChainGraph produce probabilities only
G3	Automated avoidance actions destroy digital evidence, and no framework quantifies or optimises evidence preservation	Autonomous defence research treats evidence as irrelevant
G4	Predictions are not grounded in verifiable evidence, limiting analyst trust and auditability	Most GNN and LSTM predictors are black-box models
G5	High reported accuracy comes from complete benchmarks; robustness under missing or erased evidence is untested	Evaluations rely on clean DARPA and CIC datasets

6. Proposed Novelty: Evidence-Preserving Pre-emptive Containment (EPPC)

EPPC is a framework that predicts the attacker’s next ATT&CK stage from digital evidence and selects the avoidance action that interrupts that stage while causing the least loss of forensic evidence.

6.1 Novelty Components

ID	Contribution	What Is New	Gap Addressed
N1	Evidence Loss Cost (ELC) Model: each response action is scored by the forensic artefacts it destroys (network isolation = low, process termination = high, host re-imaging = maximum)	First quantified link between response actions and evidence loss	G3
N2	Dual-Objective Action Selection: choose action a minimising $\text{Risk}(\text{predicted stage} \mid a) + \lambda \cdot \text{ELC}(a)$, with λ set by organisational policy	Joint optimisation of prevention and forensic readiness	G1, G2
N3	Evidence-Grounded Prediction: each prediction must cite its supporting provenance subgraph; low-support predictions are deferred rather than acted upon	Auditable, action-safe prediction	G4
N4	Evidence-Degradation Robustness Protocol: evaluation with 10–50% of logs removed to simulate log clearing and sensor failure	Realistic evaluation absent from prior studies	G5

6.2 Novelty Statement

Unlike existing systems that either investigate attacks after the fact or block them without regard to forensic consequences, the proposed framework predicts the next attack stage from digital evidence and selects containment actions that jointly minimise attack progression and forensic evidence loss.

6.3 Evaluation Metrics

Metric	Definition	Status
Next-Stage Top-k Accuracy	Proportion of cases where the true next stage is among the top k predictions	Standard
Attack Interruption Rate	Percentage of attacks stopped before Exfiltration or Impact	Standard
Prediction Lead Time	Time between prediction and actual occurrence of the stage	Standard
Evidence Retention Score (ERS)	Fraction of forensically relevant artefacts preserved after response	Newly proposed
Degradation Robustness	Performance drop as evidence completeness decreases	Newly proposed protocol

7. Proposed Methodology

Stage	Method	Tools / Resources
Datasets	Host provenance and APT scenarios; network-level supplement	DARPA OpTC, DARPA Transparent Computing E3/E5, ATLAS dataset, CIC-IDS2018
Evidence Graph Construction	Parse audit events into temporal provenance graphs	Python, NetworkX, PyTorch Geometric
Technique Mapping	Label subgraphs with ATT&CK tactics and techniques	MITRE ATT&CK STIX data
Stage Prediction	Temporal Graph Network combined with Transformer encoder	PyTorch, PyTorch Geometric Temporal

Stage	Method	Tools / Resources
Avoidance Decision	ELC table with cost-sensitive optimiser; optional RL agent	Custom optimiser, CybORG
Explainability	Extract minimal supporting evidence subgraph	GNNExplainer
Baselines	GNN+LSTM stage estimator, ensemble predictor, block-all SOAR policy	Re-implemented baselines

8. Expected Outcomes

- A unified framework linking forensic evidence analysis, attack prediction and pre-emptive avoidance.
- A reusable Evidence Loss Cost table for common incident response actions.
- Improved evidence retention compared with conventional automated response, at comparable attack interruption rates.
- Explainable, evidence-backed predictions suitable for analyst review.
- A publishable contribution targeting IEEE Access, Elsevier Computers & Security or Forensic Science International: Digital Investigation.

9. Conclusion

Research between 2020 and 2026 has advanced forensic investigation, attack-stage prediction and autonomous response considerably, yet these streams remain disconnected, and the forensic cost of defensive action has not been addressed. By treating evidence preservation as an explicit optimisation objective alongside attack prevention, the proposed EPPC framework offers a distinct and measurable contribution. Before final submission, a targeted search for terms such as “evidence-preserving incident response” and “forensic-aware automated response” is recommended to confirm that no closely related study has appeared.

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